

2nd Mar 2026 | DAG401

⇒ Adam Optimizer

- Adam (Adaptive Moment Estimation)
 - ↳ momentum (first moment)
 - ↳ variance (second moment)

- Momentum

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

exponential moving avg. of gradients

provides direcⁿ & velocity

$$\beta_1 = 0.9$$

m_t
↓
where
should we
go?

- Variance (Adaptive LR)

$$v_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

EMA of squared gradients

controls step size per param.

$$\beta_2 = 0.999$$

how big
our step
must
be?

$$- m_0 = 0, v_0 = 0$$

$$m_1 = (1 - \beta_1) g_1$$

$$v_1 = (1 - \beta_2) g_1^2$$

$$E[m_t] = E[g_t] \cdot (1 - \beta_1^t)$$

$$E[v_t] = E[g_t^2] \cdot (1 - \beta_2^t)$$

- by correcting the bias -
 - moments are correctly estimated
 - stable initial training
 - better convergence

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

- for updating params -

$$\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad \epsilon = 10^{-8}$$

⇒ NAdam

Nesterov acc. adam

try for best of both worlds (Adam gives adaptive learning rate & nesterov gives better momentum) → CV & less budget intensive

faster convergence than Adam

- std. momentum

$$v_t = \beta v_{t-1} + (1 - \beta) v_t$$

$$\theta_{t+1} = \theta_t - \alpha v_t$$

- Nesterov momentum

$$\bar{\theta}_t = \theta_t + \beta v_{t-1}$$

$$g_t = \nabla J(\bar{\theta}_t)$$

$$v_t = \beta v_{t-1} + (1 - \beta) g_t$$

$$\theta_{t+1} = \theta_t - \alpha v_t$$

- Nesterov evaluates gradient at the anticipated posⁿ, allowing for better correction when it overshoots. Works good for convex optimizaⁿ.

- Nesterov modificaⁿ looks like in NAdam

$$\bar{m}_t = \beta_1 \hat{m}_t + \frac{(1 - \beta_1) g_t}{1 - \beta_1^t}$$

$$\theta_{t+1} = \theta_t - \alpha \frac{\bar{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$